

CS 402 Project Proposal

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1 Overview

With the rapid growth in technology and worsening work-life balance in today's times, people have started to neglect their physical health and dietary habits. At the same time, there has been a massive rise in the use of gamified systems, and individuals are increasingly gravitating towards their smartphones for personal organization and habit tracking.

AURA (Activity, Utility, Routine, Analytics) is a single-page application (SPA) that addresses this by helping users improve their fitness, tracking workouts, and meals in one centralized place. To capitalize on this gamified mobile trend, users earn "Aura points" for consistency, streaks, and goal completion. This system supports multiple user roles, secure sessions, server-side pagination/filtering, and third-party API integration. The application will be built using the MEAN stack and deployed on the departmental OpenStack system.

2 User Stories

This application will have two roles: Standard User and Administrator.

2.1 Standard User

As a standard User, I want to be able to:

- Log in with my username and password, and log out securely.
- Edit my profile information, such as setting my profile visibility to public or private.
- Filter my workout history by muscle group to view detailed progress over time
- Search for dishes to retrieve nutritional data, routed entirely through the server
- Add retrieved nutritional items to my daily food log to track my macros
- Keep my workout and nutrition logs strictly private, preventing other users from accessing my data unless explicitly shared

- Earn "aura points" for completing workouts

2.2 Administrator

As an Administrator, I want to be able to:

- Log in securely with my credentials and log out
- Edit my administrative profile
- Create a new user account, and enable or disable existing accounts
- Change any user's password, including other Admins
- View a server-side sorted list of all registered users to effectively manage the platform's 5,000+ accounts

3 Data Models

The application will manage four primary entities based on the proposed architecture: **USER**, **EXERCISE**, **MEAL**, and **WORKOUT**.

Example of table reference: see **Table ??**. Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Entity	Field	Type	Description	Req?	Indexed?
USER	_id	ObjectId	Primary Key	Yes	Yes (primary)
	username	String	NOT NULL, Unique	Yes	Yes (unique)
	email	String	NOT NULL, Unique	Yes	Yes (unique)
	passwordHash	String	NOT NULL	Yes	No
	role	Enum	User role	Yes	Yes
	auraPoints	Number	Gamification score	Yes	Yes
	heightCm	Number	Height in cm	No	No
	weightKg	Number	Current weight in kg	No	No
	goalWeightKg	Number	Target weight	No	No
	age	Number	User's age	No	No
	sex	String	Biological sex	No	No
EXERCISE	exerciseId	Uuid	Primary Key	Yes	Yes (primary)
	name	String	Name of movement	Yes	Yes (text)
	muscleGroup	String	Targeted muscle	Yes	Yes
	equipment	String	Required equipment	Yes	Yes
	instructions	String	Form details	No	No
	imageUrl	String	Demo image link	No	No
MEAL	mealId	Uuid	Primary Key	Yes	Yes (primary)
	userId	Uuid	Foreign Key (Owner)	Yes	Yes
	foodName	String	Food Name	Yes	Yes (text index)
	calories	Number	Total energy content	Yes	No
	protein	Number	Protein content (g)	Yes	No
	carbs	Number	Carb content (g)	Yes	No
	fat	Number	Fat content (g)	Yes	Yes
	date	Date	Date of consumption	Yes	Yes
WORKOUT	workoutId	Uuid	Primary Key	Yes	Yes (primary)
	userId	Uuid	Foreign Key (Owner)	Yes	Yes
	exerciseId	Uuid	Foreign Key (Movement)	Yes	Yes
	name	String	Name of the session	Yes	Yes (text index)
	duration	Number	Duration in minutes	Yes	No
	caloriesBurned	Number	Energy expenditure	Yes	No
	isPublic	Boolean	Privacy flag	Yes	Yes
	createdAt	Date	Timestamp of session log	Yes	Yes

4 Third Party API

API Providers: USDA FoodData Central API (Nutrition) and ExerciseDB API (Fitness)

Data Elements Used:

Usage in Application: To satisfy the project requirements, the client application will never communicate directly with either external API. When a user searches for a meal or a new exercise to add to their routine, the client sends a request to our custom Node.js REST API. Our server will then securely query the USDA or ExerciseDB API using our private backend keys. The server will parse the returned JSON data, persist the relevant information in our MongoDB database, and finally serve that formatted data back to the client.

5 Prototype Screenshots

Link to Figma Prototype: <https://www.figma.com/proto/9MnOxr2C8nvmdZRte5y8i3/Figma-basics?node-id=654-9&t=hYSK1KmuF6aT5fsH-1>

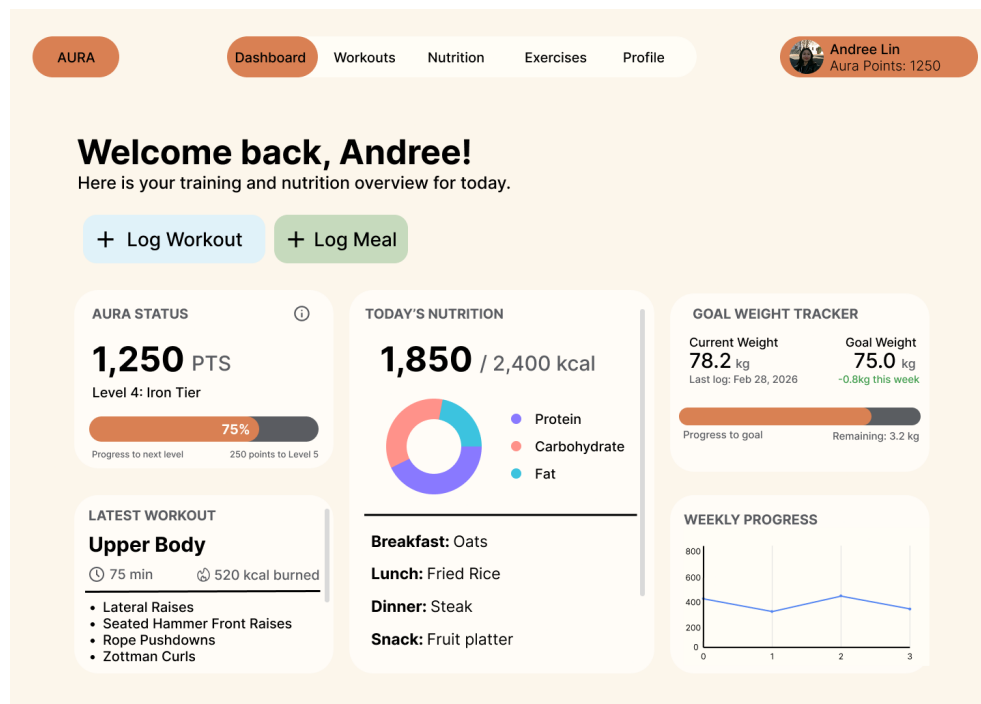
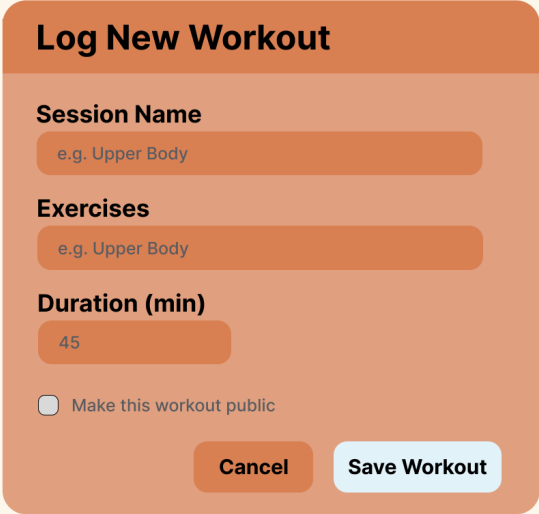


Figure 1: Dashboard



The 'Log New Workout' form is a rounded rectangle with an orange header and a light orange body. It contains three text input fields with placeholder text 'e.g. Upper Body', a duration input field with the value '45', and a checkbox labeled 'Make this workout public'. At the bottom are 'Cancel' and 'Save Workout' buttons.

Log New Workout

Session Name
e.g. Upper Body

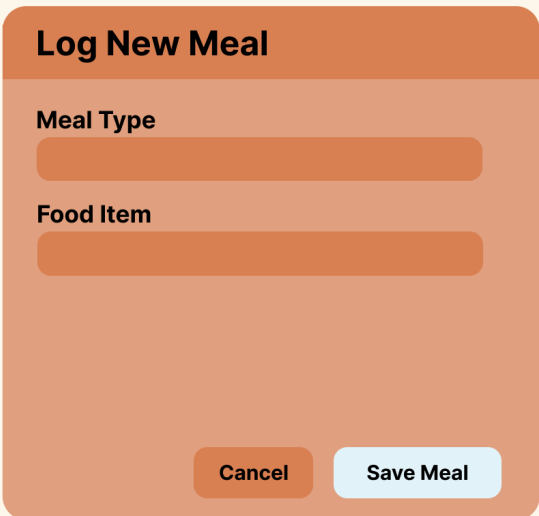
Exercises
e.g. Upper Body

Duration (min)
45

☐ Make this workout public

Cancel **Save Workout**

Figure 2: Logging a workout



The 'Log New Meal' form is a rounded rectangle with an orange header and a light orange body. It contains two text input fields, one for 'Meal Type' and one for 'Food Item'. At the bottom are 'Cancel' and 'Save Meal' buttons.

Log New Meal

Meal Type

Food Item

Cancel **Save Meal**

Figure 3: Logging a meal

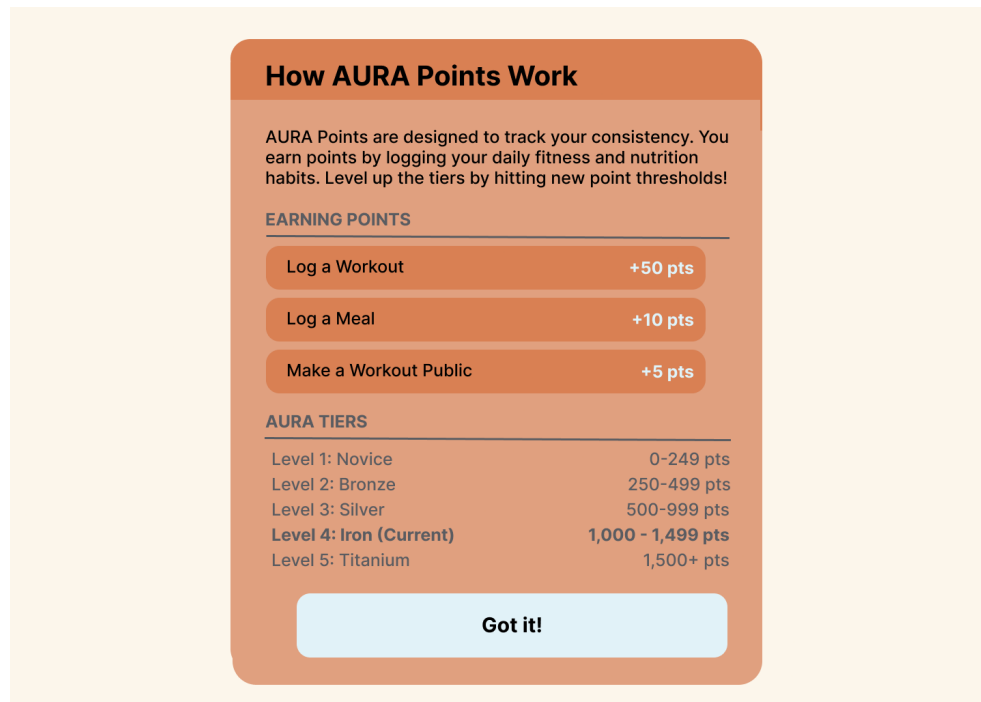


Figure 4: Information explaining AURA Points

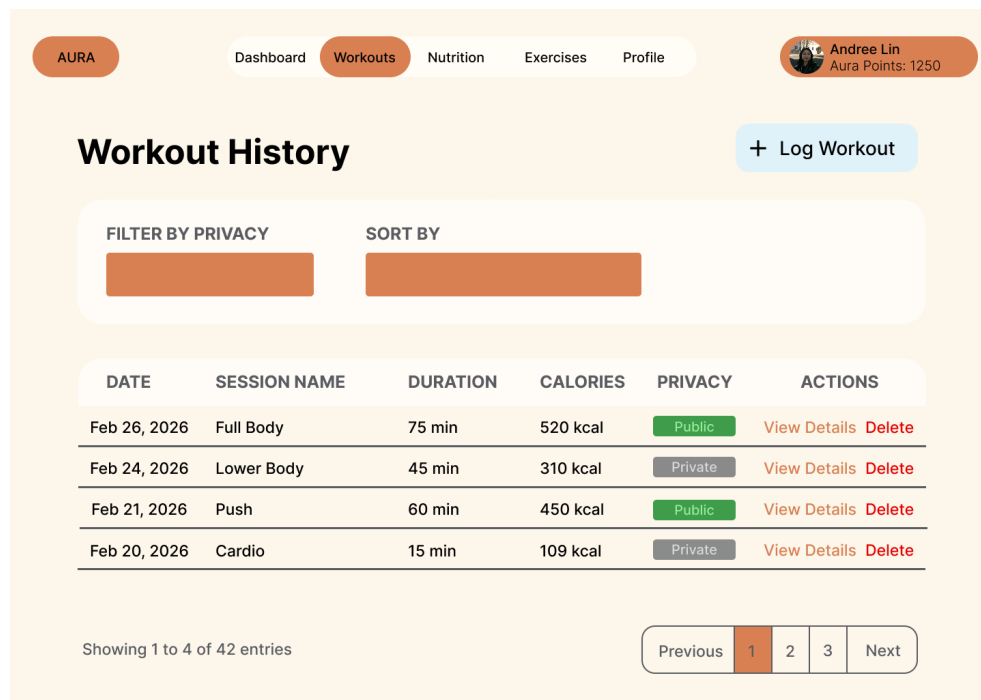


Figure 5: Workouts page

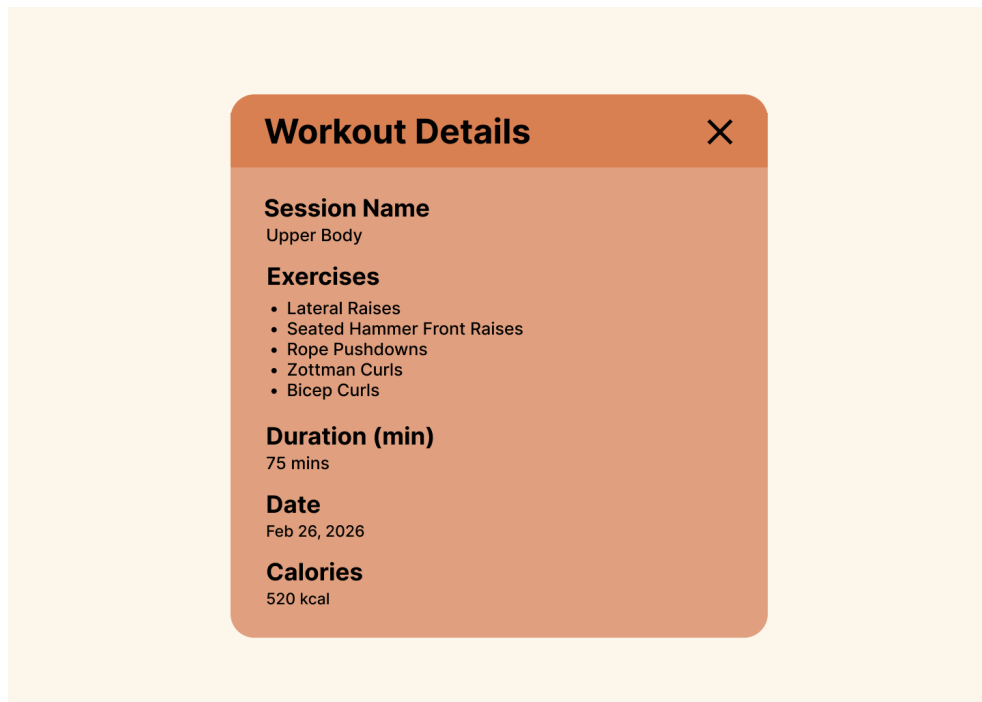


Figure 6: Extra details about each workout

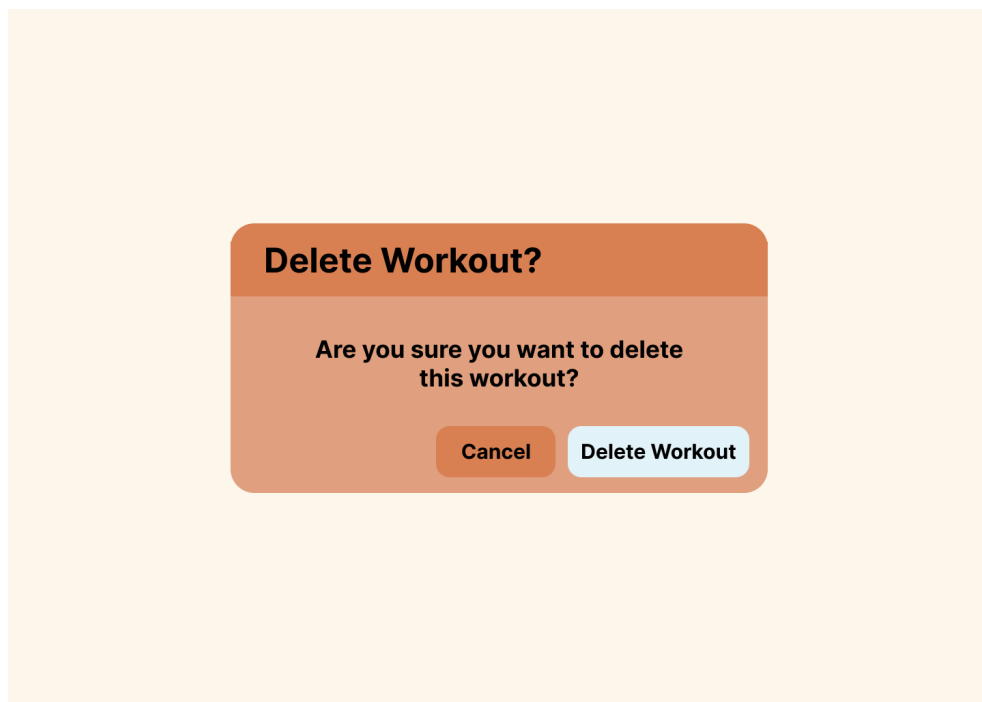


Figure 7: Delete workout confirmation

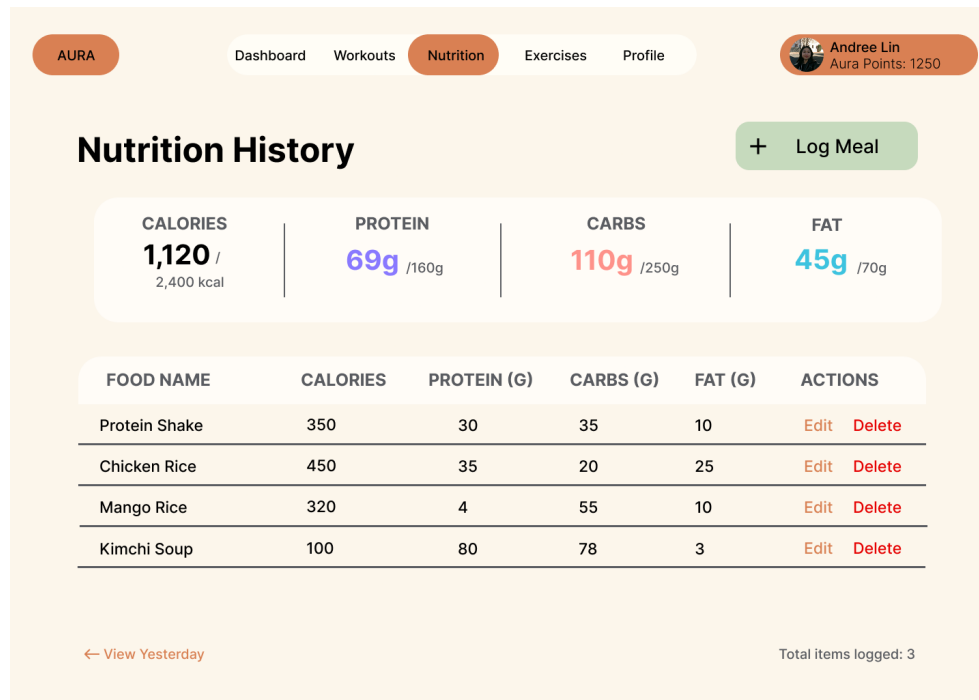


Figure 8: Nutrition Page

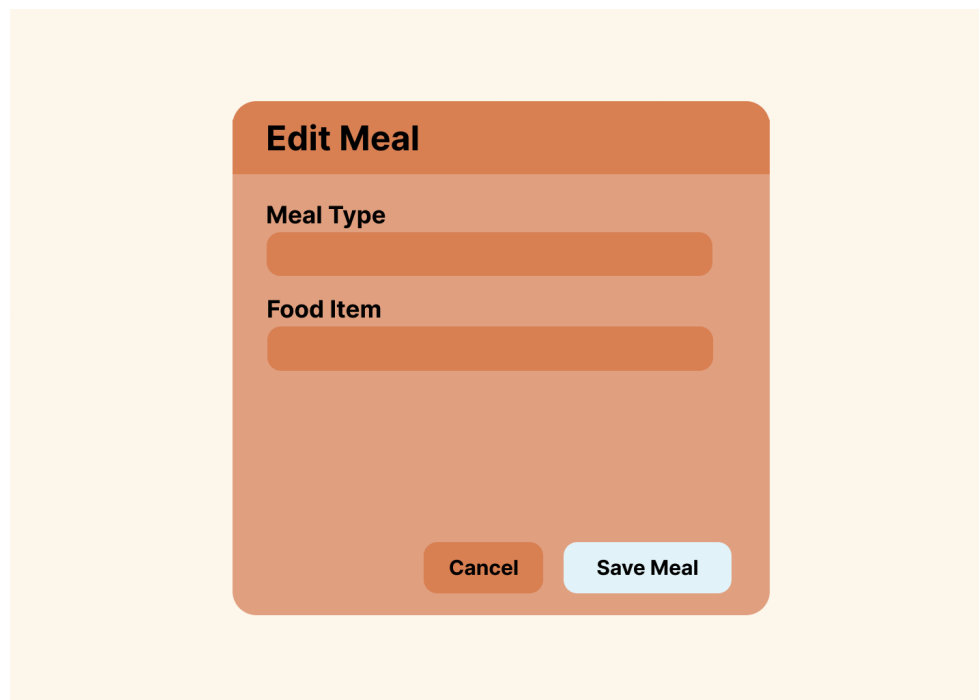


Figure 9: Edit Meal Pop-up

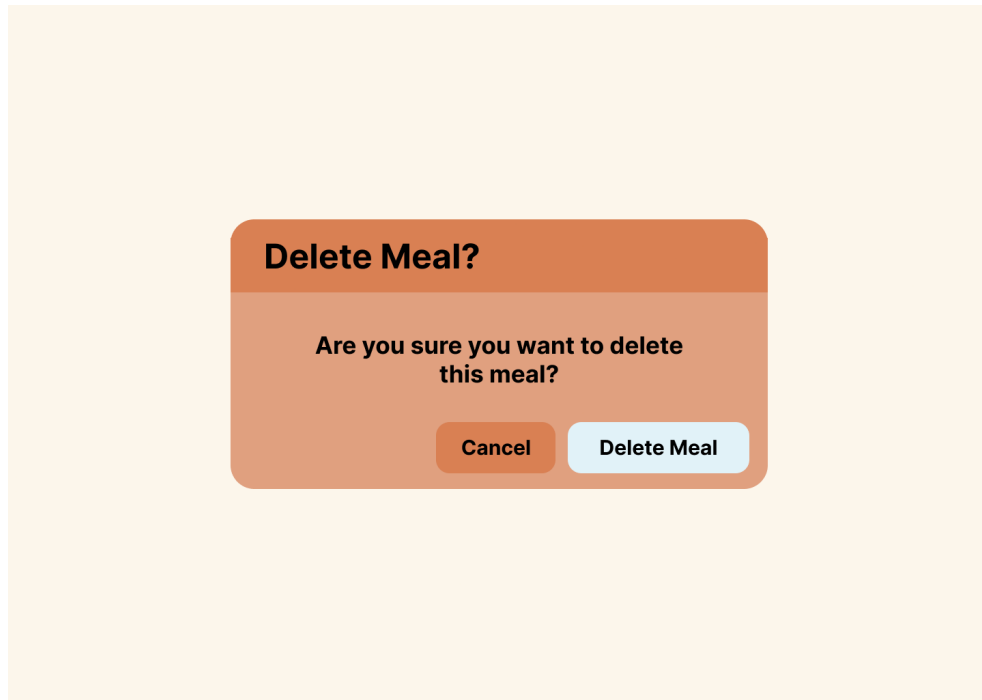


Figure 10: Delete meal confirmation

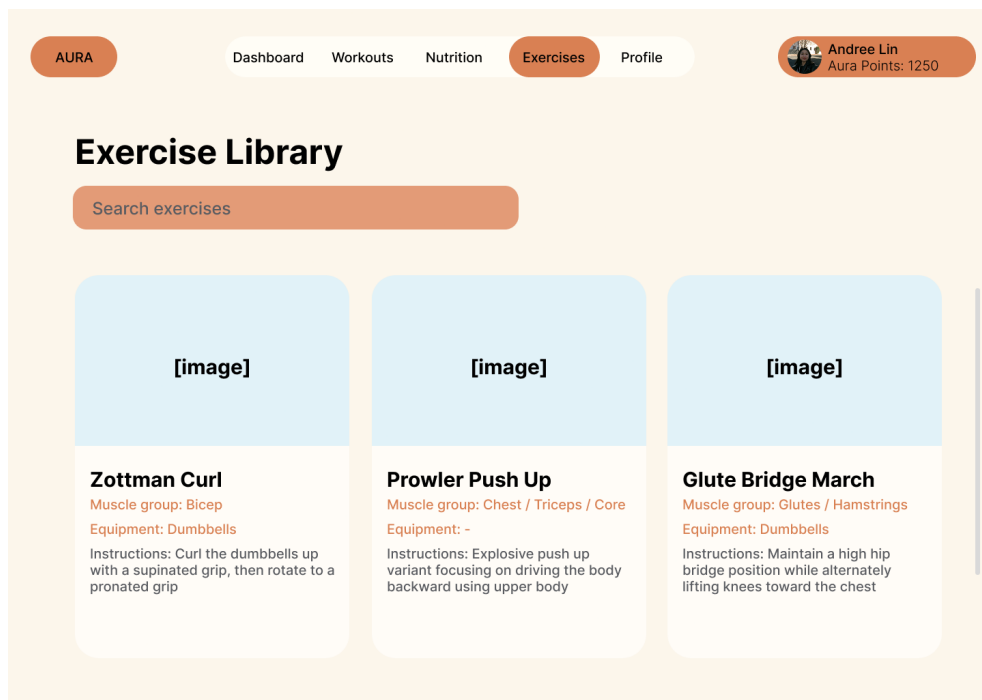


Figure 11: Library to search for exercises

AURA

DashboardWorkoutsNutritionExercisesProfile

Andree Lin

Aura Points: 1250



Profile Settings

Manage your account details and body metrics.

ACCOUNT DETAILS



Edit profile picture

Username

AndreeLin

Email Address

lin5940@uwlax.edu

Role

Admin

[Change Password](#)

BODY METRICS & GOALS

Full Name

Andree Lin

Age

22

Biological Sex

Female

Height (cm)

175

Current Weight

78.2

Goal Weight

75.0

Cancel

Save Profile

Figure 12: Viewing and editing profile page

Password Change

Old Password

New Password

New Password again

Cancel

Confirm

Figure 13: Changing password confirmation

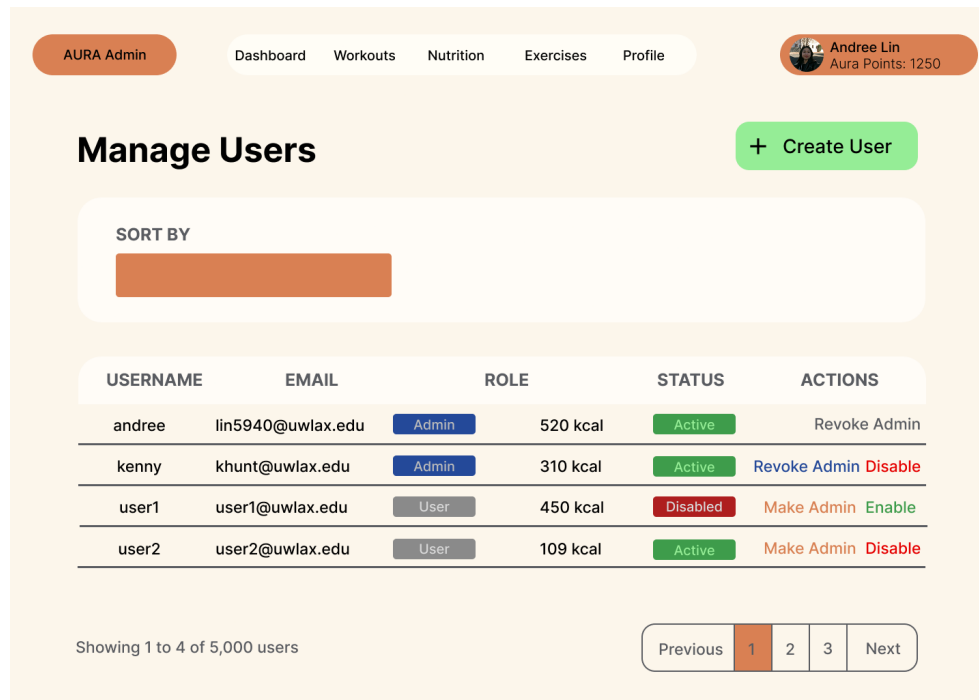


Figure 14: Admin panel - accessed by clicking top left

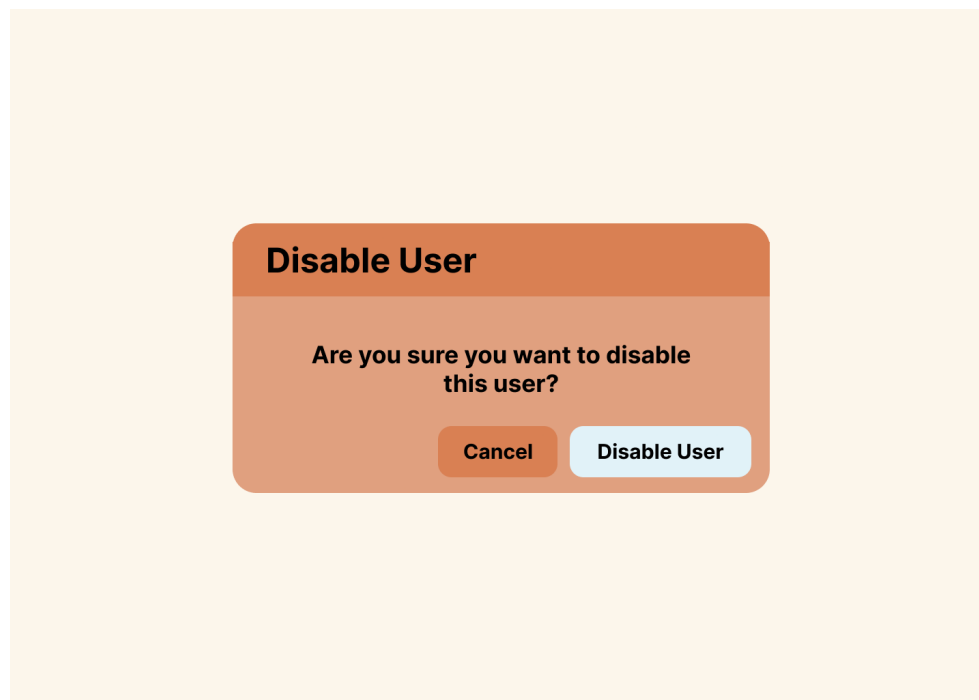


Figure 15: Disabling user confirmation

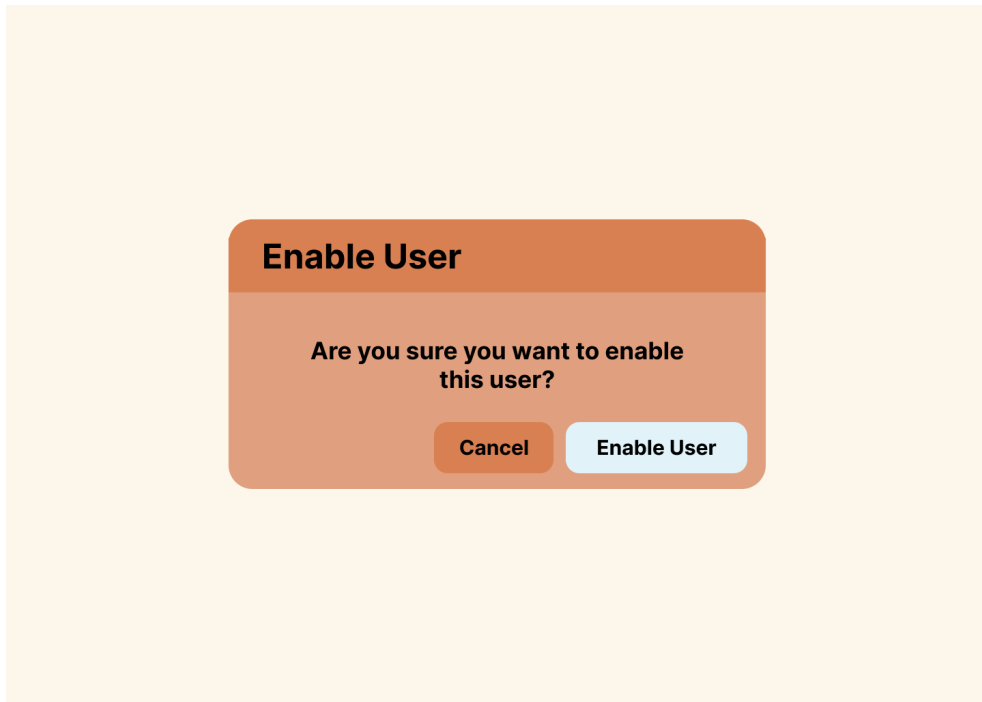


Figure 16: Enabling user confirmation

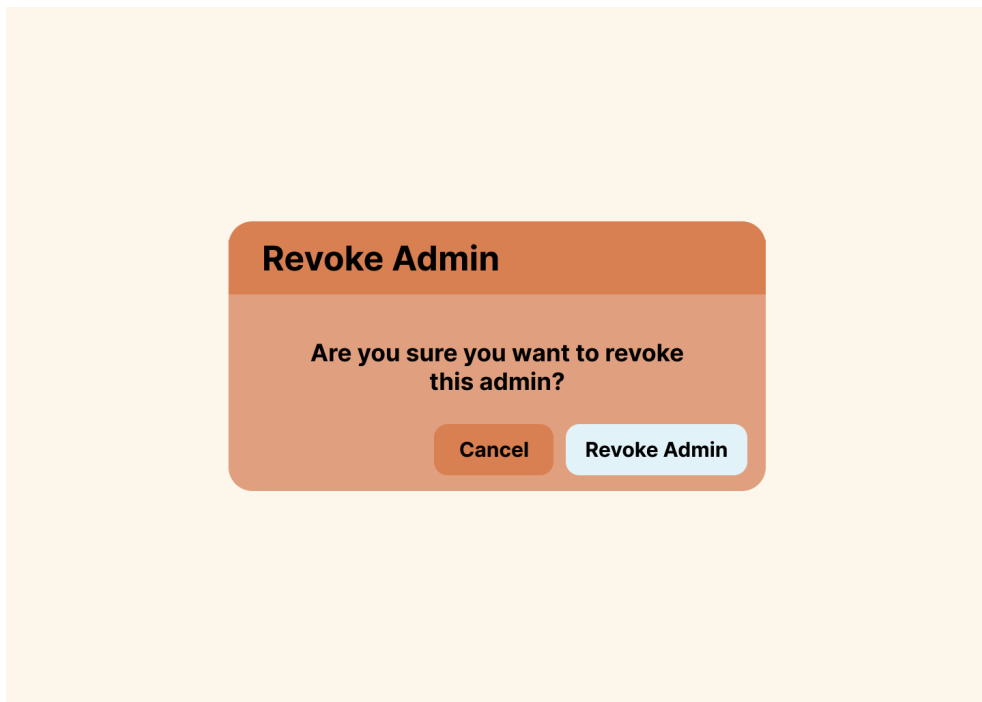


Figure 17: Revoking admin confirmation

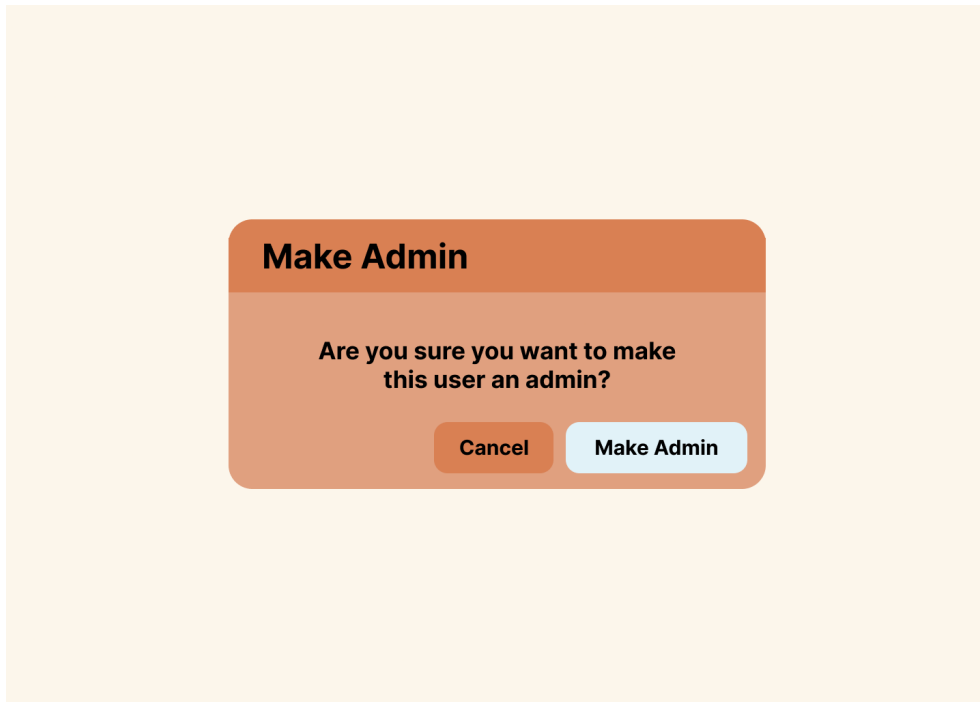


Figure 18: Making a user an admin confirmation

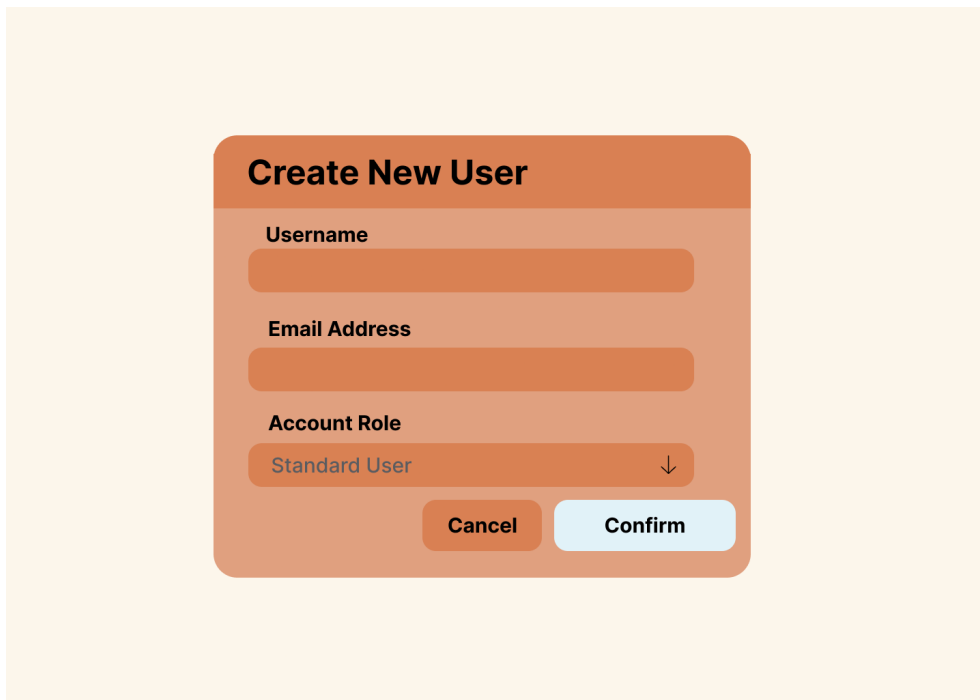


Figure 19: Creating user pop-up